Reef Disturbance, Habitat Structure, and Biodiversity Across Coastal Reef Sites

Damselfish (Pomacentridae)

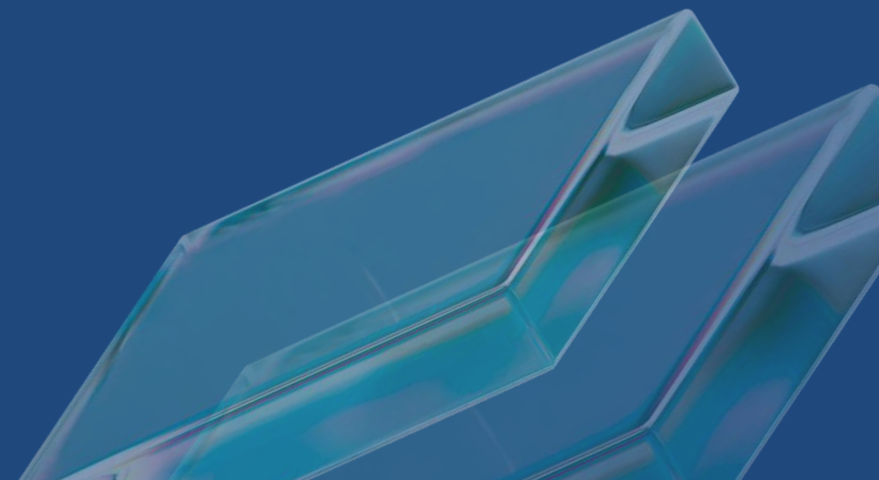
Ecology Biotutorial: Allen Coral Atlas + WorldPop + GBIF

Brent B. Al-os
Vanessa B. Bairoy
Althea C. Baldado
Angel Rose G. Bermejo
Dirk Infante
Mishelle Ann Jabel
Mikylla Jamin
Antonio Ellazhar Q. Lizada
Rachelle A. Quebec



HYPOTHESIS AND DESIGN

- **Hypothesis**: damselfish richness is higher at sites with more coral/algae habitat
- 10 sites, ranked 1 (lowest expected disturbance) to 10 (highest), each with a 20-km radius
- Habitat: Allen Coral Atlas | Species records: GBIF (rgbif), Pomacentridae



Study sites and expected disturbance

Ten coastal sites in southern Negros Oriental, ranked from 1 (low) to 10 (high).

Rank	Site	Reason for expected disturbance
1	Apo Island	Protected reef baseline
2	Dauin Poblacion	Light local human influence
3	Masaplod Norte	Moderate-low coastal development
4	Masaplod Sur	Growing community footprint
5	Mainit	Intermediate human pressure
6	Luca	Moderate human activity
7	Dauin Coastal Reef	Localized shoreline development
8	Zamboanguita Reef	Elevated human activity
9	Maluay	Runoff and population pressure
10	Salag MPA	Southern coastal comparison

Reef habitat composition

Site	Coral/algae (%)	Seagrass (%)	Reef area (km²)
Apo Island	79.52	9.14	0.9509
Dauin Poblacion	86.74	5.32	0.1415
Masaplod Norte	81.71	16.71	0.2012
Masaplod Sur	66.67	32.71	0.1329
Mainit	84.07	15.65	0.2422
Luca	100.00	Not reported	0.0242
Dauin Coastal Reef	80.05	19.95	0.1175
Zamboanguita Reef	78.81	7.84	0.2274
Maluay	97.19	2.31	0.1452
Salag MPA	93.22	3.23	0.2157

Reef topography across the sites

Reef slope and reef flat are mapped geomorphic categories.

Site	Reef slope (%)	Reef flat (%)
Apo Island	67.04	27.00
Dauin Poblacion	80.59	19.42
Masaplod Norte	70.63	28.28
Masaplod Sur	47.75	51.59
Mainit	70.53	28.27
Luca	100.00	Not reported
Dauin Coastal Reef	9.64	29.68
Zamboanguita Reef	34.48	58.66
Maluay	Not reported	69.69
Salag MPA	88.48	11.25

Results & Analysis



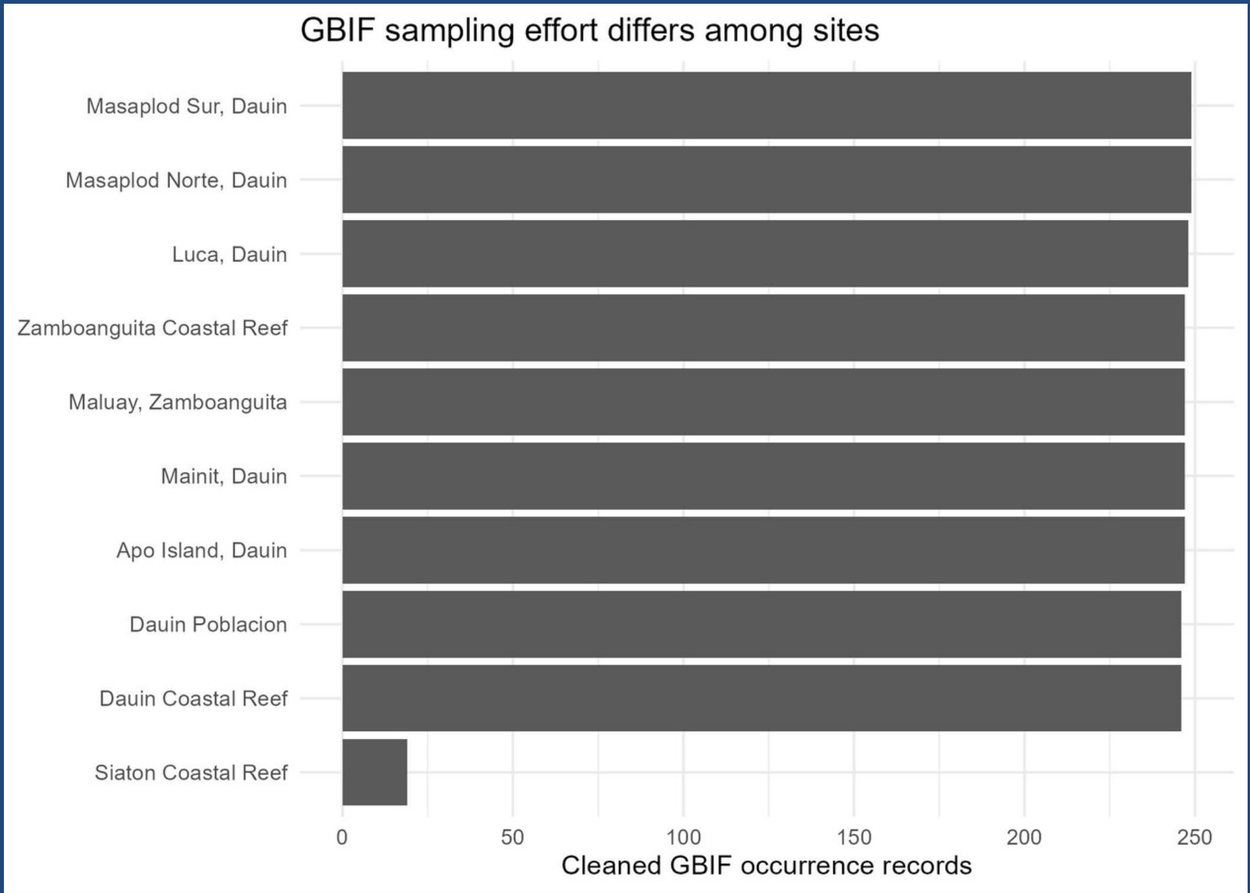


Figure 1: GBIF_RECORDS_BY_SITE

This graph shows the number of cleaned GBIF damselfish records within 20 km of each site. Nine sites had 839–874 records, while Salag MPA, Siaton had only 20, showing that the amount of available data differed greatly at this site.

Process:

- Downloaded GBIF records per site using a 20-km radius
- Cleaned the data
- Grouped the records by site and counted them
- Plotted the totals as a bar graph

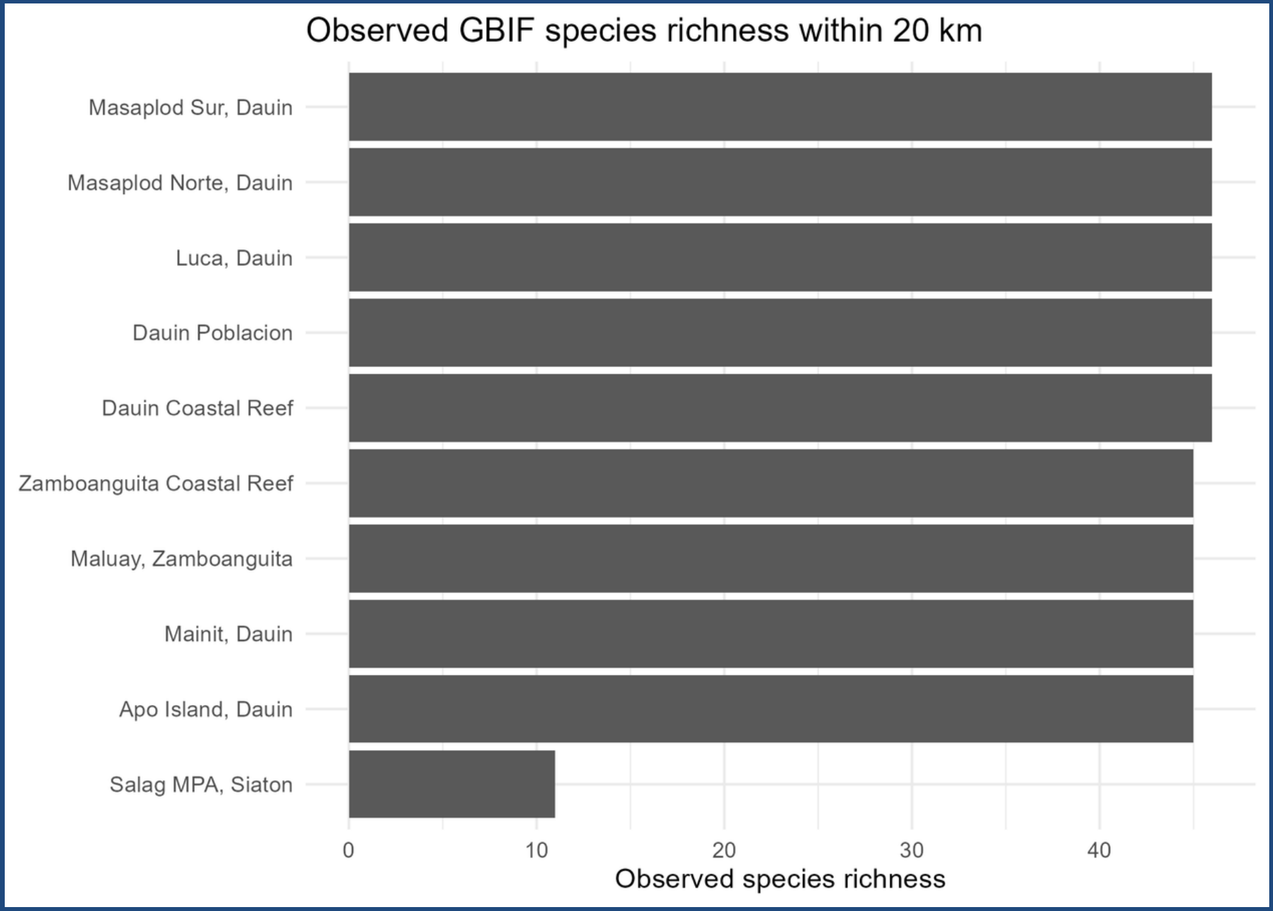


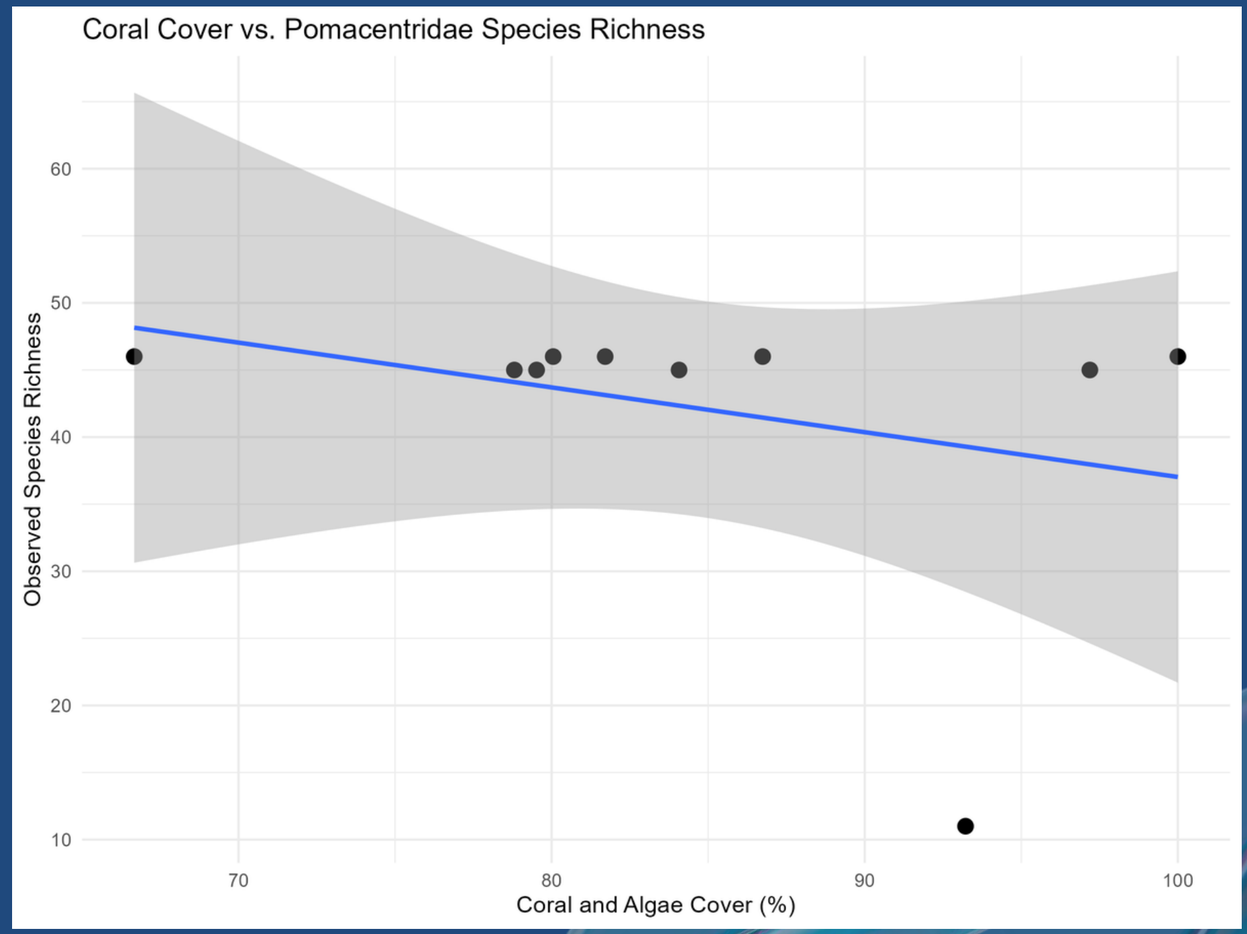
Figure 2. Observed species richness of within a 20-km radius of each site.

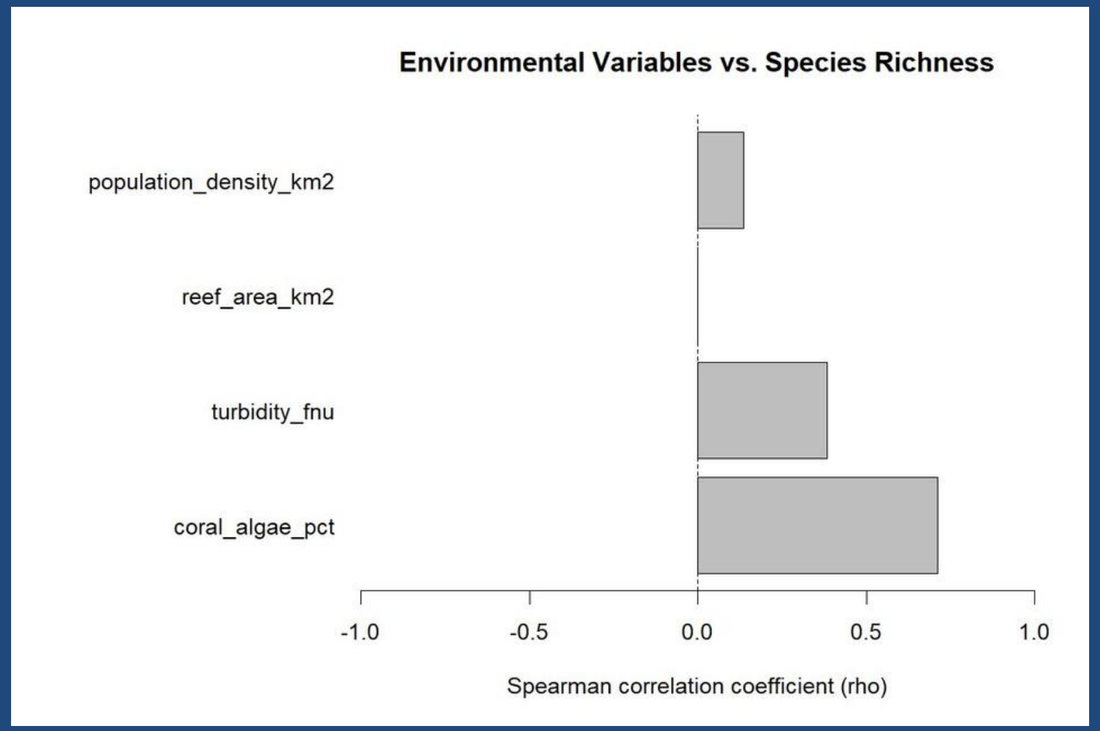
This graph shows the observed number of damselfish species at each of the 10 sites. Dauin Poblacion, Masaplod Norte, Masaplod Sur, Luca, and Dauin Coastal Reef had the highest richness with 46 species each, four sites had 45, and Salag MPA, Siaton had the lowest with 11.

Process:

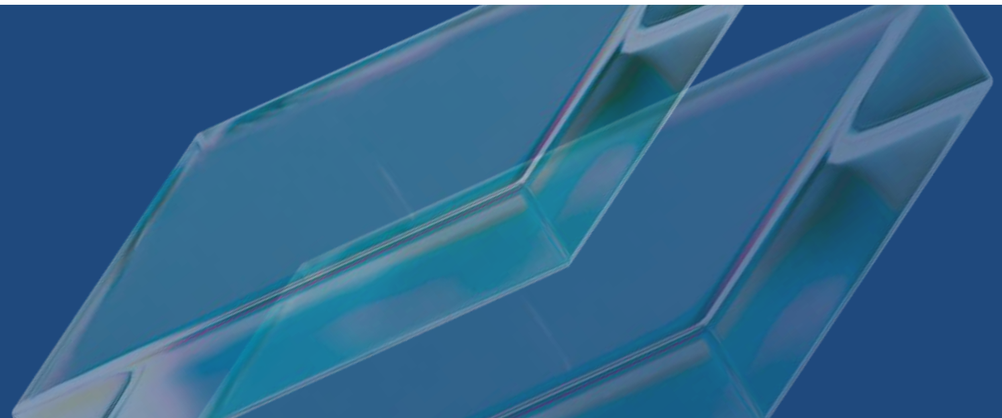
- grouped the records by site and species
- counted the distinct species per site
- created a bar graph.

Coral/Algae Cover vs. Damselfish Richness





variable	rho	p_value
coral_algae_pct	0.7100648373	0.02140516911
turbidity_fnu	0.3820465887	0.275944
reef_area_km2	0	1
population_densi	0.1348399725	0.7103422619



Summary

Variable	Spearman_rho	P_value	Interpretation	Data_type
coral_algae_pct	0.71	0.021	Positive associat	Practice analysis
turbidity_fnu	0.382	0.276	Not statistically s	Practice analysis
reef_area_km2	0	1	No rank associat	Practice analysis
population_densi	0.135	0.71	Not statistically s	Practice analysis

GBIF records and observed richness

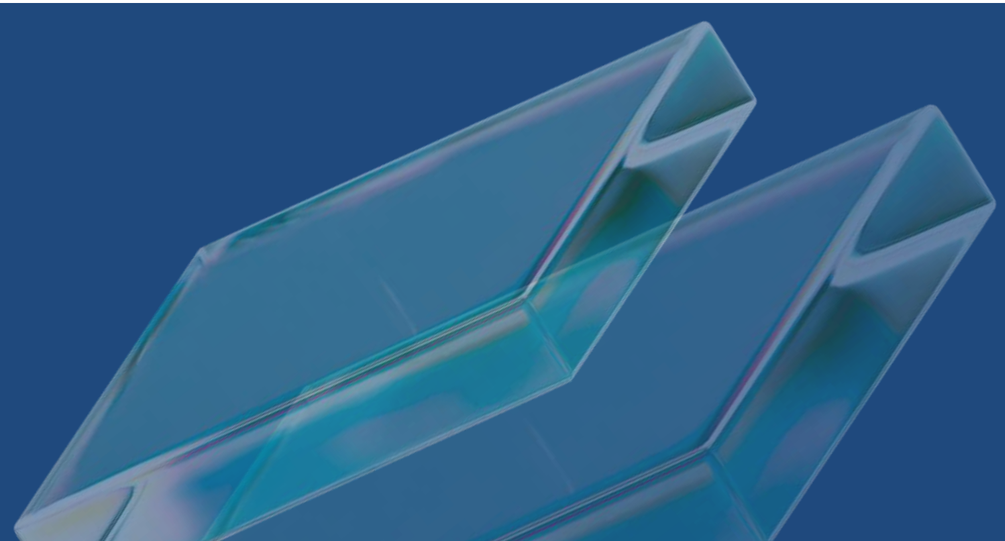
All values are based on cleaned occurrence records within each 20-km buffer.

Site	Cleaned GBIF records	Observed species
Apo Island	848	45
Dauin Poblacion	859	46
Masaplod Norte	866	46
Masaplod Sur	874	46
Mainit	848	45
Luca	865	46
Dauin Coastal Reef	859	46
Zamboanguita Reef	848	45
Maluay	839	45
Salag MPA	20	11

Correlation results

Spearman rank correlations with observed damselfish species richness.

Variable	Spearman ρ	p-value	n
Coral/algae cover	−0.114	0.754	10
Seagrass cover	0.648	0.059	9
Rock cover	0.013	0.976	8
Reef slope	−0.037	0.924	9
Reef flat	0.183	0.638	9
Lagoon	−0.339	0.411	8
Disturbance rank	−0.482	0.158	10



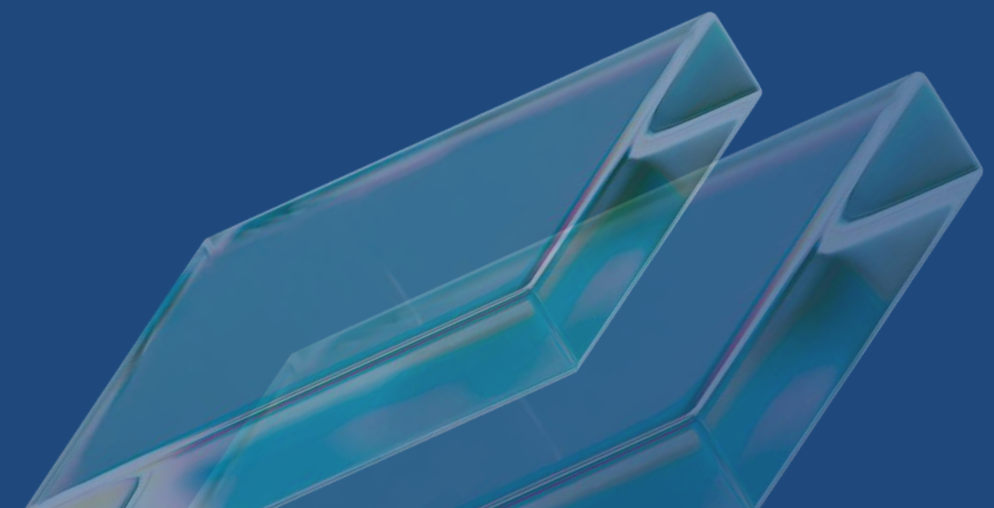
Regression result

Quasi-Poisson GLM: richness ~ coral/algae cover

Predictor	Coefficient	Standard error	p-value	Sites
Coral/algae (%)	−0.0785	0.0906	0.411	10

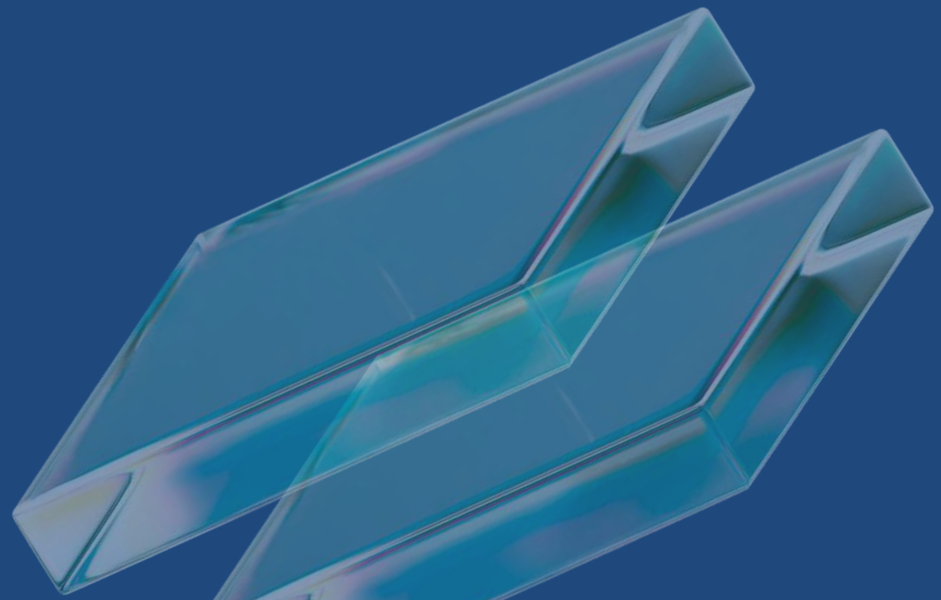
The estimated relationship is negative, but it is not statistically significant.

This model does not provide evidence that higher mapped coral/algae cover predicts higher observed richness.



Results suggest

- Nine sites had similar observed richness (45–46 species); Salag MPA had 11.
- Coral/algae cover did not show a significant relationship with richness.
- Seagrass showed a positive trend, but the p-value was slightly above 0.05.



Limitations and conclusion

Limitations

- Nearby 20-km buffers overlap, so sites are not fully independent.
- GBIF records reflect observation effort as well as species occurrence.
- Turbidity and population density are missing from the environmental file.
- Habitat class areas and expected disturbance ranks need validation.

Conclusion

The available results do not show a significant coral/algae–richness relationship. More balanced sampling and verified environmental data are needed.

Data sources

- Allen Coral Atlas – benthic habitat and geomorphic classifications
- GBIF – Pomacentridae occurrence records (2007–2026 in downloaded dataset)
- WorldPop – named in the site-ranking notes; population values not yet available
- Group R outputs – site_summary.csv, site_environment_filled.csv, correlation_results.csv, glm_results_summary.csv

Thank You

